

38. Prove the sum of angles formula for the sine function by following these steps. Fix  $x$ .

- (a) Let  $f(t) := \sin(x+t)$ . Show that  $f''(t) + f(t) = 0$ ,  $f(0) = \sin x$ , and  $f'(0) = \cos x$ .  
 (b) Use the auxiliary equation technique to solve the initial value problem  $y'' + y = 0$ ,  $y(0) = \sin x$ , and  $y'(0) = \cos x$ .

- (c) By uniqueness, the solution in part (b) is the same as  $f(t)$  from part (a). Write this equality; this should be the standard sum of angles formula for  $\sin(x+t)$ .

## 4.4 Nonhomogeneous Equations: the Method of Undetermined Coefficients

In this section we employ “judicious guessing” to derive a simple procedure for finding a solution to a *nonhomogeneous* linear equation with constant coefficients

$$(1) \quad ay'' + by' + cy = f(t),$$

when the nonhomogeneity  $f(t)$  is a single term of a special type. Our experience in Section 4.3 indicates that (1) will have an infinite number of solutions. For the moment we are content to find one particular solution. To motivate the procedure, let's first look at a few instructive examples.

**Example 1** Find a particular solution to

$$(2) \quad y'' + 3y' + 2y = 3t.$$

**Solution** We need to find a function  $y(t)$  such that the combination  $y'' + 3y' + 2y$  is a linear function of  $t$ —namely,  $3t$ . Now what kind of function  $y$  “ends up” as a linear function after having its zeroth, first, and second derivatives combined? One immediate answer is: *another linear function*. So we might try  $y_1(t) = At$  and attempt to match up  $y_1'' + 3y_1' + 2y_1$  with  $3t$ .

Perhaps you can see that this won't work:  $y_1 = At$ ,  $y_1' = A$  and  $y_1'' = 0$  gives us

$$y_1'' + 3y_1' + 2y_1 = 3A + 2At,$$

and for this to equal  $3t$ , we require both that  $A = 0$  and  $A = 3/2$ . We'll have better luck if we append a constant term to the trial function:  $y_2(t) = At + B$ . Then  $y_2' = A$ ,  $y_2'' = 0$ , and

$$y_2'' + 3y_2' + 2y_2 = 3A + 2(At + B) = 2At + (3A + 2B),$$

which successfully matches up with  $3t$  if  $2A = 3$  and  $3A + 2B = 0$ . Solving this system gives  $A = 3/2$  and  $B = -9/4$ . Thus, the function

$$y_2(t) = \frac{3}{2}t - \frac{9}{4}$$

is a solution to (2). ♦

Example 1 suggests the following method for finding a particular solution to the equation

$$ay'' + by' + cy = Ct^m, \quad m = 0, 1, 2, \dots;$$

namely, we guess a solution of the form

$$y_p(t) = A_mt^m + \dots + A_1t + A_0,$$

with undetermined coefficients  $A_j$ , and match the corresponding powers of  $t$  in  $ay'' + by' + cy$  with  $Ct^m$ .<sup>†</sup> This procedure involves solving  $m+1$  linear equations in the  $m+1$  unknowns

<sup>†</sup>In this case the coefficient of  $t^k$  in  $ay'' + by' + cy$  will be zero for  $k \neq m$  and  $C$  for  $k = m$ .

$A_0, A_1, \dots, A_m$ , and hopefully they have a solution. The technique is called the **method of undetermined coefficients**. Note that, as Example 1 demonstrates, we must retain *all* the powers  $t^m, t^{m-1}, \dots, t^1, t^0$  in the trial solution even though they are not present in the nonhomogeneity  $f(t)$ .

**Example 2** Find a particular solution to

$$(3) \quad y'' + 3y' + 2y = 10e^{3t}.$$

**Solution** We guess  $y_p(t) = Ae^{3t}$  because then  $y'_p$  and  $y''_p$  will retain the same exponential form:

$$y''_p + 3y'_p + 2y_p = 9Ae^{3t} + 3(3Ae^{3t}) + 2(Ae^{3t}) = 20Ae^{3t}.$$

Setting  $20Ae^{3t} = 10e^{3t}$  and solving for  $A$  gives  $A = 1/2$ ; hence,

$$y_p(t) = \frac{e^{3t}}{2}$$

is a solution to (3). ♦

**Example 3** Find a particular solution to

$$(4) \quad y'' + 3y' + 2y = \sin t.$$

**Solution** Our initial action might be to guess  $y_1(t) = A \sin t$ , but this will fail because the derivatives introduce cosine terms:

$$y''_1 + 3y'_1 + 2y_1 = -A \sin t + 3A \cos t + 2A \sin t = A \sin t + 3A \cos t,$$

and matching this with  $\sin t$  would require that  $A$  equal both 1 and 0. So we include the cosine term in the trial solution:

$$\begin{aligned} y_p(t) &= A \sin t + B \cos t, \\ y'_p(t) &= A \cos t - B \sin t, \\ y''_p(t) &= -A \sin t - B \cos t, \end{aligned}$$

and (4) becomes

$$\begin{aligned} y''_p(t) + 3y'_p(t) + 2y_p(t) &= -A \sin t - B \cos t + 3A \cos t - 3B \sin t \\ &\quad + 2A \sin t + 2B \cos t \\ &= (A - 3B) \sin t + (B + 3A) \cos t \\ &= \sin t. \end{aligned}$$

The equations  $A - 3B = 1$ ,  $B + 3A = 0$  have the solution  $A = 0.1$ ,  $B = -0.3$ . Thus, the function

$$y_p(t) = 0.1 \sin t - 0.3 \cos t$$

is a particular solution to (4). ♦

More generally, for an equation of the form

$$(5) \quad ay'' + by' + cy = C \sin \beta t \quad (\text{or } C \cos \beta t),$$

the method of undetermined coefficients suggests that we guess

$$(6) \quad y_p(t) = A \cos \beta t + B \sin \beta t$$

and solve (5) for the unknowns  $A$  and  $B$ .

If we compare equation (5) with the mass–spring system equation

$$(7) \quad [\text{inertia}] \times y'' + [\text{damping}] \times y' + [\text{stiffness}] \times y = F_{\text{ext}},$$

we can interpret (5) as describing a damped oscillator, shaken with a sinusoidal force. According to our discussion in Section 4.1, then, we would expect the mass ultimately to respond by moving in synchronization with the forcing sinusoid. In other words, the form (6) is suggested by physical, as well as mathematical, experience. A complete description of forced oscillators will be given in Section 4.10.

**Example 4** Find a particular solution to

$$(8) \quad y'' + 4y = 5t^2e^t.$$

**Solution** Our experience with Example 1 suggests that we take a trial solution of the form  $y_p(t) = (At^2 + Bt + C)e^t$ , to match the nonhomogeneity in (8). We find

$$\begin{aligned} y_p &= (At^2 + Bt + C)e^t, \\ y_p' &= (2At + B)e^t + (At^2 + Bt + C)e^t, \\ y_p'' &= 2Ae^t + 2(2At + B)e^t + (At^2 + Bt + C)e^t, \\ y_p'' + 4y_p &= e^t(2A + 2B + C + 4C) + te^t(4A + B + 4B) + t^2e^t(A + 4A) \\ &= 5t^2e^t. \end{aligned}$$

Matching like terms yields  $A = 1$ ,  $B = -4/5$ , and  $C = -2/25$ . A solution is given by

$$y_p(t) = \left(t^2 - \frac{4t}{5} - \frac{2}{25}\right)e^t. \quad \blacklozenge$$

As our examples illustrate, when the nonhomogeneous term  $f(t)$  is an exponential, a sine, a cosine function, or a nonnegative integer power of  $t$  times any of these, the function  $f(t)$  itself suggests the form of a particular solution. However, certain situations thwart the straightforward application of the method of undetermined coefficients. Consider, for example, the equation

$$(9) \quad y'' + y' = 5.$$

Example 1 suggests that we guess  $y_1(t) = A$ , a zero-degree polynomial. But substitution into (9) proves futile:

$$(A)'' + (A)' = 0 \neq 5.$$

The problem arises because any constant function, such as  $y_1(t) = A$ , is a solution to the corresponding homogeneous equation  $y'' + y' = 0$ , and the undetermined coefficient  $A$  gets lost upon substitution into the equation. We would encounter the same situation if we tried to find a solution to

$$(10) \quad y'' - 6y' + 9y = e^{3t}$$

of the form  $y_1 = Ae^{3t}$ , because  $e^{3t}$  solves the associated homogeneous equation and

$$[Ae^{3t}]'' - 6[Ae^{3t}]' + 9[Ae^{3t}] = 0 \neq e^{3t}.$$

The “trick” for refining the method of undetermined coefficients in these situations smacks of the same logic as in Section 4.2, when a method was prescribed for finding second solutions to homogeneous equations with double roots. Basically, we append an extra factor of  $t$  to the trial

solution suggested by the basic procedure. In other words, to solve (9) we try  $y_p(t) = At$  instead of  $A$ :

$$\begin{aligned} (9') \quad y_p &= At, \quad y'_p = A, \quad y''_p = 0, \\ y''_p + y'_p &= 0 + A = 5, \\ A &= 5, \quad y_p(t) = 5t. \end{aligned}$$

Similarly, to solve (10) we try  $y_p = Ate^{3t}$  instead of  $Ae^{3t}$ . The trick won't work this time, because the characteristic equation of (10) has a double root and, consequently,  $Ate^{3t}$  also solves the homogeneous equation:

$$[Ate^{3t}]'' - 6[Ate^{3t}]' + 9[Ate^{3t}] = 0 \neq e^{3t}.$$

But if we append *another* factor of  $t$ ,  $y_p = At^2e^{3t}$ , we succeed in finding a particular solution:<sup>†</sup>

$$\begin{aligned} y_p &= At^2e^{3t}, \quad y'_p = 2Ate^{3t} + 3At^2e^{3t}, \quad y''_p = 2Ae^{3t} + 12Ate^{3t} + 9At^2e^{3t}, \\ y''_p - 6y'_p + 9y_p &= (2Ae^{3t} + 12Ate^{3t} + 9At^2e^{3t}) - 6(2Ate^{3t} + 3At^2e^{3t}) + 9(At^2e^{3t}) \\ &= 2Ae^{3t} = e^{3t}, \end{aligned}$$

so  $A = 1/2$  and  $y_p(t) = t^2e^{3t}/2$ .

To see why this strategy resolves the problem and to generalize it, recall the form of the original differential equation (1),  $ay'' + by' + cy = f(t)$ . Its associated auxiliary equation is

$$(11) \quad ar^2 + br + c = 0,$$

and if  $r$  is a double root, then

$$(12) \quad 2ar + b = 0$$

holds also [equation (13), Section 4.2, page 163].

Now suppose the nonhomogeneity  $f(t)$  has the form  $Ct^ne^{rt}$ , and we seek to match this  $f(t)$  by substituting  $y_p(t) = (A_nt^n + A_{n-1}t^{n-1} + \cdots + A_1t + A_0)e^{rt}$  into (1), *with the power  $n$  to be determined*. For simplicity we merely list the leading terms in  $y_p$ ,  $y'_p$ , and  $y''_p$ :

$$\begin{aligned} y_p &= A_nt^n e^{rt} + A_{n-1}t^{n-1} e^{rt} + A_{n-2}t^{n-2} e^{rt} + (\text{lower-order terms}) \\ y'_p &= A_n r t^n e^{rt} + A_n n t^{n-1} e^{rt} + A_{n-1} r t^{n-1} e^{rt} + A_{n-1} (n-1) t^{n-2} e^{rt} \\ &\quad + A_{n-2} r t^{n-2} e^{rt} + (\text{l.o.t.}), \\ y''_p &= A_n r^2 t^n e^{rt} + 2A_n n r t^{n-1} e^{rt} + A_n n(n-1) t^{n-2} e^{rt} \\ &\quad + A_{n-1} r^2 t^{n-1} e^{rt} + 2A_{n-1} r(n-1) t^{n-2} e^{rt} + A_{n-2} r^2 t^{n-2} e^{rt} + (\text{l.o.t.}). \end{aligned}$$

Then the left-hand member of (1) becomes

$$\begin{aligned} (13) \quad ay''_p + by'_p + cy_p &= A_n(ar^2 + br + c)t^n e^{rt} + [A_n n(2ar + b) + A_{n-1}(ar^2 + br + c)]t^{n-1} e^{rt} \\ &\quad + [A_n n(n-1)a + A_{n-1}(n-1)(2ar + b) + A_{n-2}(ar^2 + br + c)]t^{n-2} e^{rt} \\ &\quad + (\text{l.o.t.}), \end{aligned}$$

<sup>†</sup>Indeed, the solution  $t^2$  to the equation  $y'' = 2$ , computed by simple integration, can also be derived by appending two factors of  $t$  to the solution  $y \equiv 1$  of the associated homogeneous equation.

and we observe the following:

**Case 1.** If  $r$  is *not* a root of the auxiliary equation, the leading term in (13) is  $A_n(ar^2 + br + c)t^n e^{rt}$ , and to match  $f(t) = Ct^m e^{rt}$  we must take  $n = m$ :

$$y_p(t) = (A_m t^m + \cdots + A_1 t + A_0) e^{rt}.$$

**Case 2.** If  $r$  is a *simple* root of the auxiliary equation, (11) holds and the leading term in (13) is  $A_n n(2ar + b)t^{n-1} e^{rt}$ , and to match  $f(t) = Ct^m e^{rt}$  we must take  $n = m + 1$ :

$$y_p(t) = (A_{m+1} t^{m+1} + A_m t^m + \cdots + A_1 t + A_0) e^{rt}.$$

However, now the final term  $A_0 e^{rt}$  can be dropped, since it solves the associated homogeneous equation, so we can factor out  $t$  and for simplicity renumber the coefficients to write

$$y_p(t) = t(A_m t^m + \cdots + A_1 t + A_0) e^{rt}.$$

**Case 3.** If  $r$  is a *double* root of the auxiliary equation, (11) and (12) hold and the leading term in (13) is  $A_n n(n-1)at^{n-2} e^{rt}$ , and to match  $f(t) = Ct^m e^{rt}$  we must take  $n = m + 2$ :

$$y_p(t) = (A_{m+2} t^{m+2} + A_{m+1} t^{m+1} + \cdots + A_2 t^2 + A_1 t + A_0) e^{rt},$$

but again we drop the solutions to the associated homogeneous equation and renumber to write

$$y_p(t) = t^2(A_m t^m + \cdots + A_1 t + A_0) e^{rt}.$$

We summarize with the following rule.

### Method of Undetermined Coefficients

To find a particular solution to the differential equation

$$ay'' + by' + cy = Ct^m e^{rt},$$

where  $m$  is a nonnegative integer, use the form

$$(14) \quad y_p(t) = t^s (A_m t^m + \cdots + A_1 t + A_0) e^{rt},$$

with

- (i)  $s = 0$  if  $r$  is not a root of the associated auxiliary equation;
- (ii)  $s = 1$  if  $r$  is a simple root of the associated auxiliary equation; and
- (iii)  $s = 2$  if  $r$  is a double root of the associated auxiliary equation.

To find a particular solution to the differential equation

$$ay'' + by' + cy = \begin{cases} Ct^m e^{\alpha t} \cos \beta t \\ \text{or} \\ Ct^m e^{\alpha t} \sin \beta t \end{cases}$$

for  $\beta \neq 0$ , use the form

$$(15) \quad y_p(t) = t^s (A_m t^m + \cdots + A_1 t + A_0) e^{\alpha t} \cos \beta t \\ + t^s (B_m t^m + \cdots + B_1 t + B_0) e^{\alpha t} \sin \beta t,$$

with

- (iv)  $s = 0$  if  $\alpha + i\beta$  is not a root of the associated auxiliary equation; and
- (v)  $s = 1$  if  $\alpha + i\beta$  is a root of the associated auxiliary equation.

[The  $(\cos, \sin)$  formulation (15) is easily derived from the exponential formulation (14) by putting  $r = \alpha + i\beta$  and employing Euler's formula, as in Section 4.3.]

**Remark 1.** The nonhomogeneity  $Ct^m$  corresponds to the case when  $r = 0$ .

**Remark 2.** The rigorous justification of the method of undetermined coefficients [including the analysis of the terms we dropped in (13)] will be presented in a more general context in Chapter 6.

**Example 5** Find the form for a particular solution to

$$(16) \quad y'' + 2y' - 3y = f(t),$$

where  $f(t)$  equals

$$(a) 7 \cos 3t \quad (b) 2te^t \sin t \quad (c) t^2 \cos \pi t \quad (d) 5e^{-3t} \quad (e) 3te^t \quad (f) t^2 e^t$$

**Solution** The auxiliary equation for the homogeneous equation corresponding to (16),  $r^2 + 2r - 3 = 0$ , has roots  $r_1 = 1$  and  $r_2 = -3$ . Notice that the functions in (a), (b), and (c) are associated with *complex* roots (because of the trigonometric factors). These are clearly different from  $r_1$  and  $r_2$ , so the solution forms correspond to (15) with  $s = 0$ :

$$(a) \quad y_p(t) = A \cos 3t + B \sin 3t$$

$$(b) \quad y_p(t) = (A_1 t + A_0) e^t \cos t + (B_1 t + B_0) e^t \sin t$$

$$(c) \quad y_p(t) = (A_2 t^2 + A_1 t + A_0) \cos \pi t + (B_2 t^2 + B_1 t + B_0) \sin \pi t$$

For the nonhomogeneity in (d) we appeal to (ii) and take  $y_p(t) = Ate^{-3t}$  since  $-3$  is a simple root of the auxiliary equation. Similarly, for (e) we take  $y_p(t) = t(A_1 t + A_0)e^t$  and for (f) we take  $y_p(t) = t(A_2 t^2 + A_1 t + A_0)e^t$ . ♦

**Example 6** Find the form of a particular solution to

$$y'' - 2y' + y = f(t),$$

for the same set of nonhomogeneities  $f(t)$  as in Example 5.

**Solution** Now the auxiliary equation for the corresponding homogeneous equation is  $r^2 - 2r + 1 = (r - 1)^2 = 0$ , with the double root  $r = 1$ . This root is not linked with any of the nonhomogeneities (a) through (d), so the same trial forms should be used for (a), (b), and (c) as in the previous example, and  $y(t) = Ae^{-3t}$  will work for (d).

Since  $r = 1$  is a double root, we have  $s = 2$  in (14) and the trial forms for (e) and (f) have to be changed to

$$(e) \quad y_p(t) = t^2(A_1 t + A_0)e^t$$

$$(f) \quad y_p(t) = t^2(A_2 t^2 + A_1 t + A_0)e^t$$

respectively, in accordance with (iii). ♦

**Example 7** Find the form of a particular solution to

$$y'' - 2y' + 2y = 5te^t \cos t.$$

**Solution** Now the auxiliary equation for the corresponding homogeneous equation is  $r^2 - 2r + 2 = 0$ , and it has complex roots  $r_1 = 1 + i$ ,  $r_2 = 1 - i$ . Since the nonhomogeneity involves  $e^{\alpha t} \cos \beta t$  with  $\alpha = \beta = 1$ ; that is,  $\alpha + i\beta = 1 + i = r_1$ , the solution takes the form

$$y_p(t) = t(A_1 t + A_0)e^t \cos t + t(B_1 t + B_0)e^t \sin t. \quad \blacklozenge$$

The nonhomogeneity  $\tan t$  in an equation like  $y'' + y' + y = \tan t$  is not one of the forms for which the method of undetermined coefficients can be used; the derivatives of the “trial solution”  $y(t) = A \tan t$ , for example, get complicated, and it is not clear what additional terms need to be added to obtain a true solution. In Section 4.6 we discuss a different procedure that can handle such nonhomogeneous terms. **Keep in mind that the method of undetermined coefficients applies only to nonhomogeneities that are polynomials, exponentials, sines or cosines, or products of these functions.** The superposition principle in Section 4.5 shows how the method can be extended to the **sums** of such nonhomogeneities. Also, it provides the key to assembling a general solution to (1) that can accommodate initial value problems, which we have avoided so far in our examples.

## 4.4 EXERCISES

In Problems 1–8, decide whether or not the method of undetermined coefficients can be applied to find a particular solution of the given equation.

1.  $y'' + 2y' - y = t^{-1}e^t$
2.  $5y'' - 3y' + 2y = t^3 \cos 4t$
3.  $2y''(x) - 6y'(x) + y(x) = (\sin x)/e^{4x}$
4.  $x'' + 5x' - 3x = 3^t$
5.  $y''(\theta) + 3y'(\theta) - y(\theta) = \sec \theta$
6.  $2\omega''(x) - 3\omega(x) = 4x \sin^2 x + 4x \cos^2 x$
7.  $8z'(x) - 2z(x) = 3x^{100}e^{4x} \cos 25x$
8.  $ty'' - y' + 2y = \sin 3t$

In Problems 9–26, find a particular solution to the differential equation.

9.  $y'' + 3y = -9$
10.  $y'' + 2y' - y = 10$
11.  $y''(x) + y(x) = 2^x$
12.  $2x' + x = 3t^2$
13.  $y'' - y' + 9y = 3 \sin 3t$
14.  $2z'' + z = 9e^{2t}$
15.  $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = xe^x$
16.  $\theta''(t) - \theta(t) = t \sin t$
17.  $y'' + 4y = 8 \sin 2t$
18.  $y'' - 2y' + y = 8e^t$
19.  $4y'' + 11y' - 3y = -2te^{-3t}$
20.  $y'' + 4y = 16t \sin 2t$
21.  $x''(t) - 4x'(t) + 4x(t) = te^{2t}$

22.  $x''(t) - 2x'(t) + x(t) = 24t^2e^t$
23.  $y''(\theta) - 7y'(\theta) = \theta^2$
24.  $y''(x) + y(x) = 4x \cos x$
25.  $y'' + 2y' + 4y = 111e^{2t} \cos 3t$
26.  $y'' + 2y' + 2y = 4te^{-t} \cos t$

In Problems 27–32, determine the form of a particular solution for the differential equation. (Do not evaluate coefficients.)

27.  $y'' + 9y = 4t^3 \sin 3t$
28.  $y'' - 6y' + 9y = 5t^6e^{3t}$
29.  $y'' + 3y' - 7y = t^4e^t$
30.  $y'' - 2y' + y = 7e^t \cos t$
31.  $y'' + 2y' + 2y = 8t^3e^{-t} \sin t$
32.  $y'' - y' - 12y = 2t^6e^{-3t}$

In Problems 33–36, use the method of undetermined coefficients to find a particular solution to the given higher-order equation.

33.  $y''' - y'' + y = \sin t$
34.  $2y''' + 3y'' + y' - 4y = e^{-t}$
35.  $y''' + y'' - 2y = te^t$
36.  $y^{(4)} - 3y'' - 8y = \sin t$

## 4.5 The Superposition Principle and Undetermined Coefficients Revisited

The next theorem describes the superposition principle, a very simple observation which nonetheless endows the solution set for our equations with a powerful structure. It extends the applicability of the method of undetermined coefficients and enables us to solve initial value problems for nonhomogeneous differential equations.

### Superposition Principle

**Theorem 3.** If  $y_1$  is a solution to the differential equation

$$ay'' + by' + cy = f_1(t) ,$$

and  $y_2$  is a solution to

$$ay'' + by' + cy = f_2(t) ,$$

then for any constants  $k_1$  and  $k_2$ , the function  $k_1y_1 + k_2y_2$  is a solution to the differential equation

$$ay'' + by' + cy = k_1f_1(t) + k_2f_2(t) .$$

**Proof.** This is straightforward; by substituting and rearranging we find

$$\begin{aligned} a(k_1y_1 + k_2y_2)'' + b(k_1y_1 + k_2y_2)' + c(k_1y_1 + k_2y_2) \\ = k_1(ay_1'' + by_1' + cy_1) + k_2(ay_2'' + by_2' + cy_2) \\ = k_1f_1(t) + k_2f_2(t) . \quad \blacklozenge \end{aligned}$$

**Example 1** Find a particular solution to

$$(1) \quad y'' + 3y' + 2y = 3t + 10e^{3t} \quad \text{and}$$

$$(2) \quad y'' + 3y' + 2y = -9t + 20e^{3t} .$$

**Solution** In Example 1, Section 4.4, we found that  $y_1(t) = 3t/2 - 9/4$  was a solution to  $y'' + 3y' + 2y = 3t$ , and in Example 2 we found that  $y_2(t) = e^{3t}/2$  solved  $y'' + 3y' + 2y = 10e^{3t}$ . By superposition, then,  $y_1 + y_2 = 3t/2 - 9/4 + e^{3t}/2$  solves equation (1).

The right-hand member of (2) equals minus three times  $(3t)$  plus two times  $(10e^{3t})$ . Therefore, this same combination of  $y_1$  and  $y_2$  will solve (2):

$$y(t) = -3y_1 + 2y_2 = -3(3t/2 - 9/4) + 2(e^{3t}/2) = -9t/2 + 27/4 + e^{3t} . \quad \blacklozenge$$

If we take a particular solution  $y_p$  to a nonhomogeneous equation like

$$(3) \quad ay'' + by' + cy = f(t)$$

and add it to a general solution  $c_1y_1 + c_2y_2$  of the homogeneous equation associated with (3),

$$(4) \quad ay'' + by' + cy = 0 ,$$

the sum

$$(5) \quad y(t) = y_p(t) + c_1y_1(t) + c_2y_2(t)$$

is again, according to the superposition principle, a solution to (3):

$$\begin{aligned} a(y_p + c_1y_1 + c_2y_2)'' + b(y_p + c_1y_1 + c_2y_2)' + c(y_p + c_1y_1 + c_2y_2) \\ = f(t) + 0 + 0 = f(t) . \end{aligned}$$

Since (5) contains two parameters, one would suspect that  $c_1$  and  $c_2$  can be chosen to make it satisfy arbitrary initial conditions. It is easy to verify that this is indeed the case.

### Existence and Uniqueness: Nonhomogeneous Case

**Theorem 4.** For any real numbers  $a (\neq 0)$ ,  $b$ ,  $c$ ,  $t_0$ ,  $Y_0$ , and  $Y_1$ , suppose  $y_p(t)$  is a particular solution to (3) in an interval  $I$  containing  $t_0$  and that  $y_1(t)$  and  $y_2(t)$  are linearly independent solutions to the associated homogeneous equation (4) in  $I$ . Then there exists a unique solution in  $I$  to the initial value problem

$$(6) \quad ay'' + by' + cy = f(t), \quad y(t_0) = Y_0, \quad y'(t_0) = Y_1,$$

and it is given by (5), for the appropriate choice of the constants  $c_1$ ,  $c_2$ .

**Proof.** We have already seen that the superposition principle implies that (5) solves the differential equation. To satisfy the initial conditions in (6) we need to choose the constants so that

$$(7) \quad \begin{cases} y_p(t_0) + c_1y_1(t_0) + c_2y_2(t_0) = Y_0, \\ y_p'(t_0) + c_1y_1'(t_0) + c_2y_2'(t_0) = Y_1. \end{cases}$$

But as in the proof of Theorem 2 in Section 4.2, simple algebra shows that the choice

$$c_1 = \frac{[Y_0 - y_p(t_0)]y_2'(t_0) - [Y_1 - y_p'(t_0)]y_2(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)} \quad \text{and} \\ c_2 = \frac{[Y_1 - y_p'(t_0)]y_1(t_0) - [Y_0 - y_p(t_0)]y_1'(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

solves (7) unless the denominator is zero; Lemma 1, Section 4.2, assures us that it is not.

Why is the solution unique? If  $y_1(t)$  were another solution to (6), then the difference  $y_{II}(t) := y_p(t) + c_1y_1(t) + c_2y_2(t) - y_1(t)$  would satisfy

$$(8) \quad \begin{cases} ay_{II}'' + by_{II}' + cy_{II} = f(t) - f(t) = 0, \\ y_{II}(t_0) = Y_0 - Y_0 = 0, \quad y_{II}'(t_0) = Y_1 - Y_1 = 0. \end{cases}$$

But the initial value problem (8) admits the identically zero solution, and Theorem 1 in Section 4.2 applies since the differential equation in (8) is homogeneous. Consequently, (8) has *only* the identically zero solution. Thus,  $y_{II} \equiv 0$  and  $y_1 = y_p + c_1y_1 + c_2y_2$ . ♦

These deliberations entitle us to say that  $y = y_p + c_1y_1 + c_2y_2$  is a **general solution** to the nonhomogeneous equation (3), since *any* solution  $y_g(t)$  can be expressed in this form. (**Proof:** As in Section 4.2, we simply pick  $c_1$  and  $c_2$  so that  $y_p + c_1y_1 + c_2y_2$  matches the value and the derivative of  $y_g$  at *any single point*; by uniqueness,  $y_p + c_1y_1 + c_2y_2$  and  $y_g$  have to be the same function.)

**Example 2** Given that  $y_p(t) = t^2$  is a particular solution to

$$y'' - y = 2 - t^2,$$

find a general solution and a solution satisfying  $y(0) = 1$ ,  $y'(0) = 0$ .

**Solution** The corresponding homogeneous equation,

$$y'' - y = 0 ,$$

has the associated auxiliary equation  $r^2 - 1 = 0$ . Because  $r = \pm 1$  are the roots of this equation, a general solution to the homogeneous equation is  $c_1 e^t + c_2 e^{-t}$ . Combining this with the particular solution  $y_p(t) = t^2$  of the nonhomogeneous equation, we find that a general solution is

$$y(t) = t^2 + c_1 e^t + c_2 e^{-t} .$$

To meet the initial conditions, set

$$\begin{aligned} y(0) &= 0^2 + c_1 e^0 + c_2 e^{-0} = 1 , \\ y'(0) &= 2 \times 0 + c_1 e^0 - c_2 e^{-0} = 0 , \end{aligned}$$

which yields  $c_1 = c_2 = \frac{1}{2}$ . The answer is

$$y(t) = t^2 + \frac{1}{2}(e^t + e^{-t}) = t^2 + \cosh t . \quad \blacklozenge$$

**Example 3** A mass–spring system is driven by a sinusoidal external force  $(5 \sin t + 5 \cos t)$ . The mass equals 1, the spring constant equals 2, and the damping coefficient equals 2 (in appropriate units), so the deliberations of Section 4.1 imply that the motion is governed by the differential equation

$$(9) \quad y'' + 2y' + 2y = 5 \sin t + 5 \cos t .$$

If the mass is initially located at  $y(0) = 1$ , with a velocity  $y'(0) = 2$ , find its equation of motion.

**Solution** The associated homogeneous equation  $y'' + 2y' + 2y = 0$  was studied in Example 1, Section 4.3; the roots of the auxiliary equation were found to be  $-1 \pm i$ , leading to a general solution  $c_1 e^{-t} \cos t + c_2 e^{-t} \sin t$ .

The method of undetermined coefficients dictates that we try to find a particular solution of the form  $A \sin t + B \cos t$  for the first nonhomogeneity  $5 \sin t$ :

$$(10) \quad \begin{aligned} y_p &= A \sin t + B \cos t , & y'_p &= A \cos t - B \sin t , & y''_p &= -A \sin t - B \cos t ; \\ y''_p + 2y'_p + 2y_p &= (-A - 2B + 2A) \sin t + (-B + 2A + 2B) \cos t = 5 \sin t . \end{aligned}$$

Matching coefficients requires  $A = 1$ ,  $B = -2$  and so  $y_p = \sin t - 2 \cos t$ .

The second nonhomogeneity  $5 \cos t$  calls for the identical form for  $y_p$  and leads to  $(-A - 2B + 2A) \sin t + (-B + 2A + 2B) \cos t = 5 \cos t$ , or  $A = 2$ ,  $B = 1$ . Hence  $y_p = 2 \sin t + \cos t$ .

By the superposition principle, a general solution to (9) is given by the sum

$$\begin{aligned} y &= c_1 e^{-t} \cos t + c_2 e^{-t} \sin t + \sin t - 2 \cos t + 2 \sin t + \cos t \\ &= c_1 e^{-t} \cos t + c_2 e^{-t} \sin t + 3 \sin t - \cos t . \end{aligned}$$

The initial conditions are

$$\begin{aligned} y(0) &= 1 = c_1 e^{-0} \cos 0 + c_2 e^{-0} \sin 0 + 3 \sin 0 - \cos 0 = c_1 - 1 , \\ y'(0) &= 2 = c_1 [-e^{-t} \cos t - e^{-t} \sin t]_{t=0} + c_2 [-e^{-t} \sin t + e^{-t} \cos t]_{t=0} \\ &\quad + 3 \cos 0 + \sin 0 \\ &= -c_1 + c_2 + 3 , \end{aligned}$$

requiring  $c_1 = 2$ ,  $c_2 = 1$ , and thus

$$(11) \quad y(t) = 2e^{-t} \cos t + e^{-t} \sin t + 3 \sin t - \cos t . \quad \blacklozenge$$

The solution (11) exemplifies the features of forced, damped oscillations that we anticipated in Section 4.1. There is a sinusoidal component  $(3 \sin t - \cos t)$  that is synchronous with the driving force  $(5 \sin t + 5 \cos t)$ , and a component  $(2e^{-t} \cos t + e^{-t} \sin t)$  that dies out. When the system is “pumped” sinusoidally, the response is a synchronous sinusoidal oscillation, after an initial transient that depends on the initial conditions; the synchronous response is the particular solution supplied by the method of undetermined coefficients, and the transient is the solution to the associated homogeneous equation. This interpretation will be discussed in detail in Sections 4.9 and 4.10.

You may have observed that, since the two undetermined-coefficient forms in the last example were identical and were destined to be added together, we could have used the form (10) to match both nonhomogeneities at the same time, deriving the condition

$$y_p'' + 2y_p' + 2y_p = (-A - 2B + 2A) \sin t + (-B + 2A + 2B) \cos t = 5 \sin t + 5 \cos t,$$

with solution  $y_p = 3 \sin t - \cos t$ . The next example illustrates this “streamlined” procedure.

**Example 4** Find a particular solution to

$$(12) \quad y'' - y = 8te^t + 2e^t.$$

**Solution** A general solution to the associated homogeneous equation is easily seen to be  $c_1 e^t + c_2 e^{-t}$ . Thus, a particular solution for matching the nonhomogeneity  $8te^t$  has the form  $t(A_1 t + A_0)e^t$ , whereas matching  $2e^t$  requires the form  $A_0 t e^t$ . Therefore, we can match both with the first form:

$$\begin{aligned} y_p &= t(A_1 t + A_0)e^t = (A_1 t^2 + A_0 t)e^t, \\ y_p' &= (A_1 t^2 + A_0 t)e^t + (2A_1 t + A_0)e^t = [A_1 t^2 + (2A_1 + A_0)t + A_0]e^t, \\ y_p'' &= [2A_1 t + (2A_1 + A_0)]e^t + [A_1 t^2 + (2A_1 + A_0)t + A_0]e^t \\ &= [A_1 t^2 + (4A_1 + A_0)t + (2A_1 + 2A_0)]e^t. \end{aligned}$$

Thus

$$\begin{aligned} y_p'' - y_p &= [4A_1 t + (2A_1 + 2A_0)]e^t \\ &= 8te^t + 2e^t, \end{aligned}$$

which yields  $A_1 = 2$ ,  $A_0 = -1$ , and so  $y_p = (2t^2 - t)e^t$ . ♦

We generalize this procedure by modifying the method of undetermined coefficients as follows.

### Method of Undetermined Coefficients (Revisited)

To find a particular solution to the differential equation

$$ay'' + by' + cy = P_m(t)e^{rt},$$

where  $P_m(t)$  is a polynomial of degree  $m$ , use the form

$$(13) \quad y_p(t) = t^s (A_m t^m + \cdots + A_1 t + A_0) e^{rt};$$

if  $r$  is not a root of the associated auxiliary equation, take  $s = 0$ ; if  $r$  is a simple root of the associated auxiliary equation, take  $s = 1$ ; and if  $r$  is a double root of the associated auxiliary equation, take  $s = 2$ .

To find a particular solution to the differential equation

$$ay'' + by' + cy = P_m(t)e^{\alpha t} \cos \beta t + Q_n(t)e^{\alpha t} \sin \beta t, \quad \beta \neq 0,$$

where  $P_m(t)$  is a polynomial of degree  $m$  and  $Q_n(t)$  is a polynomial of degree  $n$ , use the form

$$(14) \quad y_p(t) = t^s (A_k t^k + \cdots + A_1 t + A_0) e^{\alpha t} \cos \beta t \\ + t^s (B_k t^k + \cdots + B_1 t + B_0) e^{\alpha t} \sin \beta t,$$

where  $k$  is the larger of  $m$  and  $n$ . If  $\alpha + i\beta$  is not a root of the associated auxiliary equation, take  $s = 0$ ; if  $\alpha + i\beta$  is a root of the associated auxiliary equation, take  $s = 1$ .

**Example 5** Write down the form of a particular solution to the equation

$$y'' + 2y' + 2y = 5e^{-t} \sin t + 5t^3 e^{-t} \cos t.$$

**Solution** The roots of the associated homogeneous equation  $y'' + 2y' + 2y = 0$  were identified in Example 3 as  $-1 \pm i$ . Application of (14) dictates the form

$$y_p(t) = t(A_3 t^3 + A_2 t^2 + A_1 t + A_0) e^{-t} \cos t + t(B_3 t^3 + B_2 t^2 + B_1 t + B_0) e^{-t} \sin t. \quad \blacklozenge$$

The method of undetermined coefficients applies to higher-order linear differential equations with constant coefficients. Details will be provided in Chapter 6, but the following example should be clear.

**Example 6** Write down the form of a particular solution to the equation

$$y''' + 2y'' + y' = 5e^{-t} \sin t + 3 + 7te^{-t}.$$

**Solution** The auxiliary equation for the associated homogeneous is  $r^3 + 2r^2 + r = r(r+1)^2 = 0$ , with a double root  $r = -1$  and a single root  $r = 0$ . Term by term, the nonhomogeneities call for the forms

$$A_0 e^{-t} \cos t + B_0 e^{-t} \sin t \quad (\text{for } 5e^{-t} \sin t), \\ t A_0 \quad (\text{for } 3), \\ t^2 (A_1 t + A_0) e^{-t} \quad (\text{for } 7te^{-t}).$$

(If  $-1$  were a *triple* root, we would need  $t^3(A_1 t + A_0)e^{-t}$  for  $7te^{-t}$ .) Of course, we have to rename the coefficients, so the general form is

$$y_p(t) = Ae^{-t} \cos t + Be^{-t} \sin t + tC + t^2(Dt + E)e^{-t}. \quad \blacklozenge$$

## 4.5 EXERCISES

1. Given that  $y_1(t) = \cos t$  is a solution to

$$y'' - y' + y = \sin t$$

and  $y_2(t) = e^{2t}/3$  is a solution to

$$y'' - y' + y = e^{2t},$$

use the superposition principle to find solutions to the following differential equations:

(a)  $y'' - y' + y = 5 \sin t.$

(b)  $y'' - y' + y = \sin t - 3e^{2t}.$

(c)  $y'' - y' + y = 4 \sin t + 18e^{2t}.$

2. Given that  $y_1(t) = (1/4)\sin 2t$  is a solution to  $y'' + 2y' + 4y = \cos 2t$  and that  $y_2(t) = t/4 - 1/8$  is a solution to  $y'' + 2y' + 4y = t$ , use the superposition principle to find solutions to the following:

- (a)  $y'' + 2y' + 4y = t + \cos 2t$ .  
 (b)  $y'' + 2y' + 4y = 2t - 3\cos 2t$ .  
 (c)  $y'' + 2y' + 4y = 11t - 12\cos 2t$ .

In Problems 3–8, a nonhomogeneous equation and a particular solution are given. Find a general solution for the equation.

3.  $y'' - y = t$ ,  $y_p(t) = -t$   
 4.  $y'' + y' = 1$ ,  $y_p(t) = t$   
 5.  $\theta'' - \theta' - 2\theta = 1 - 2t$ ,  $\theta_p(t) = t - 1$   
 6.  $y'' + 5y' + 6y = 6x^2 + 10x + 2 + 12e^x$ ,  
 $y_p(x) = e^x + x^2$   
 7.  $y'' = 2y + 2\tan^3 x$ ,  $y_p(x) = \tan x$   
 8.  $y'' = 2y' - y + 2e^x$ ,  $y_p(x) = x^2 e^x$

In Problems 9–16 decide whether the method of undetermined coefficients together with superposition can be applied to find a particular solution of the given equation. Do not solve the equation.

9.  $3y'' + 2y' + 8y = t^2 + 4t - t^2 e^t \sin t$   
 10.  $y'' - y' + y = (e^t + t)^2$   
 11.  $y'' - 6y' - 4y = 4\sin 3t - e^{3t}t^2 + 1/t$   
 12.  $y'' + y' + ty = e^t + 7$   
 13.  $y'' - 2y' + 3y = \cosh t + \sin^3 t$   
 14.  $2y'' + 3y' - 4y = 2t + \sin^2 t + 3$   
 15.  $y'' + e^t y' + y = 7 + 3t$   
 16.  $2y'' - y' + 6y = t^2 e^{-t} \sin t - 8t \cos 3t + 10^t$

In Problems 17–22, find a general solution to the differential equation.

17.  $y'' - 2y' - 3y = 3t^2 - 5$   
 18.  $y'' - y = -11t + 1$   
 19.  $y''(x) - 3y'(x) + 2y(x) = e^x \sin x$   
 20.  $y''(\theta) + 4y(\theta) = \sin \theta - \cos \theta$   
 21.  $y''(\theta) + 2y'(\theta) + 2y(\theta) = e^{-\theta} \cos \theta$   
 22.  $y''(x) + 6y'(x) + 10y(x) = 10x^4 + 24x^3 + 2x^2 - 12x + 18$

In Problems 23–30, find the solution to the initial value problem.

23.  $y' - y = 1$ ,  $y(0) = 0$   
 24.  $y'' = 6t$ ;  $y(0) = 3$ ,  $y'(0) = -1$   
 25.  $z''(x) + z(x) = 2e^{-x}$ ;  $z(0) = 0$ ,  $z'(0) = 0$   
 26.  $y'' + 9y = 27$ ;  $y(0) = 4$ ,  $y'(0) = 6$   
 27.  $y''(x) - y'(x) - 2y(x) = \cos x - \sin 2x$ ;  
 $y(0) = -7/20$ ,  $y'(0) = 1/5$   
 28.  $y'' + y' - 12y = e^t + e^{2t} - 1$ ;  $y(0) = 1$ ,  $y'(0) = 3$

29.  $y''(\theta) - y(\theta) = \sin \theta - e^{2\theta}$ ;  
 $y(0) = 1$ ,  $y'(0) = -1$   
 30.  $y'' + 2y' + y = t^2 + 1 - e^t$ ;  $y(0) = 0$ ,  $y'(0) = 2$

In Problems 31–36, determine the form of a particular solution for the differential equation. Do not solve.

31.  $y'' + y = \sin t + t \cos t + 10^t$   
 32.  $y'' - y = e^{2t} + te^{2t} + t^2 e^{2t}$   
 33.  $x'' - x' - 2x = e^t \cos t - t^2 + \cos^3 t$   
 34.  $y'' + 5y' + 6y = \sin t - \cos 2t$   
 35.  $y'' - 4y' + 5y = e^{5t} + t \sin 3t - \cos 3t$   
 36.  $y'' - 4y' + 4y = t^2 e^{2t} - e^{2t}$

In Problems 37–40, find a particular solution to the given higher-order equation.

37.  $y''' - 2y'' - y' + 2y = 2t^2 + 4t - 9$   
 38.  $y^{(4)} - 5y'' + 4y = 10 \cos t - 20 \sin t$   
 39.  $y''' + y'' - 2y = te^t + 1$   
 40.  $y^{(4)} - 3y''' + 3y'' - y' = 6t - 20$

41. **Discontinuous Forcing Term.** In certain physical models, the nonhomogeneous term, or **forcing term**,  $g(t)$  in the equation

$$ay'' + by' + cy = g(t)$$

may not be continuous but may have a jump discontinuity. If this occurs, we can still obtain a reasonable solution using the following procedure. Consider the initial value problem

$$y'' + 2y' + 5y = g(t); \quad y(0) = 0, \quad y'(0) = 0,$$

where

$$g(t) = \begin{cases} 10 & \text{if } 0 \leq t \leq 3\pi/2 \\ 0 & \text{if } t > 3\pi/2 \end{cases}.$$

- (a) Find a solution to the initial value problem for  $0 \leq t \leq 3\pi/2$ .  
 (b) Find a general solution for  $t > 3\pi/2$ .  
 (c) Now choose the constants in the general solution from part (b) so that the solution from part (a) and the solution from part (b) agree, together with their first derivatives, at  $t = 3\pi/2$ . This gives us a continuously differentiable function that satisfies the differential equation except at  $t = 3\pi/2$ .

42. **Forced Vibrations.** As discussed in Section 4.1, a vibrating spring with damping that is under external force can be modeled by

$$(15) \quad my'' + by' + ky = g(t),$$

where  $m > 0$  is the mass of the spring system,  $b > 0$  is the damping constant,  $k > 0$  is the spring constant,  $g(t)$  is the force on the system at time  $t$ , and  $y(t)$  is the displacement from the equilibrium of the spring system at time  $t$ . Assume  $b^2 < 4mk$ .

- (a) Determine the form of the equation of motion for the spring system when  $g(t) = \sin \beta t$  by finding a general solution to equation (15).
- (b) Discuss the long-term behavior of this system. [Hint: Consider what happens to the general solution obtained in part (a) as  $t \rightarrow +\infty$ .]
43. A mass–spring system is driven by a sinusoidal external force  $g(t) = 5 \sin t$ . The mass equals 1, the spring constant equals 3, and the damping coefficient equals 4. If the mass is initially located at  $y(0) = 1/2$  and at rest, i.e.,  $y'(0) = 0$ , find its equation of motion.
44. A mass–spring system is driven by the external force  $g(t) = 2 \sin 3t + 10 \cos 3t$ . The mass equals 1, the spring constant equals 5, and the damping coefficient equals 2. If the mass is initially located at  $y(0) = -1$ , with initial velocity  $y'(0) = 5$ , find its equation of motion.

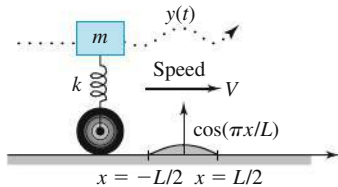


Figure 4.11 Speed bump

45. **Speed Bumps.** Often bumps like the one depicted in Figure 4.11 are built into roads to discourage speeding. The figure suggests that a crude model of the vertical motion  $y(t)$  of a car encountering the speed bump with the speed  $V$  is given by

$$y(t) = 0 \quad \text{for } t \leq -L/(2V),$$

$$my'' + ky = \begin{cases} F_0 \cos(\pi Vt/L) & \text{for } |t| < L/(2V) \\ 0 & \text{for } t \geq L/(2V). \end{cases}$$

(The absence of a damping term indicates that the car's shock absorbers are not functioning.)

- (a) Taking  $m = k = 1$ ,  $L = \pi$ , and  $F_0 = 1$  in appropriate units, solve this initial value problem. Thereby show that the formula for the oscillatory motion after the car has traversed the speed bump is  $y(t) = A \sin t$ , where the constant  $A$  depends on the speed  $V$ .



- (b) Plot the amplitude  $|A|$  of the solution  $y(t)$  found in part (a) versus the car's speed  $V$ . From the graph, estimate the speed that produces the most violent shaking of the vehicle.

46. Show that the *boundary value problem*

$$y'' + \lambda^2 y = \sin t; \quad y(0) = 0, \quad y(\pi) = 1,$$

has a solution if and only if  $\lambda \neq \pm 1, \pm 2, \pm 3, \dots$

47. Find the solution(s) to

$$y'' + 9y = 27 \cos 6t$$

(if it exists) satisfying the *boundary conditions*

(a)  $y(0) = -1, \quad y(\pi/6) = 3.$

(b)  $y(0) = -1, \quad y(\pi/3) = 5.$

(c)  $y(0) = -1, \quad y(\pi/3) = -1.$

48. All that is known concerning a mysterious second-order constant-coefficient differential equation  $y'' + py' + qy = g(t)$  is that  $t^2 + 1 + e^t \cos t$ ,  $t^2 + 1 + e^t \sin t$ , and  $t^2 + 1 + e^t \cos t + e^t \sin t$  are solutions.

- (a) Determine two linearly independent solutions to the corresponding homogeneous equation.

- (b) Find a suitable choice of  $p$ ,  $q$ , and  $g(t)$  that enables these solutions.

## 4.6 Variation of Parameters

We have seen that the method of undetermined coefficients is a simple procedure for determining a particular solution when the equation has constant coefficients and the nonhomogeneous term is of a special type. Here we present a more general method, called **variation of parameters**,<sup>†</sup> for finding a particular solution.

Consider the nonhomogeneous linear second-order equation

$$(1) \quad ay'' + by' + cy = f(t)$$

and let  $\{y_1(t), y_2(t)\}$  be two linearly independent solutions for the corresponding homogeneous equation

$$ay'' + by' + cy = 0.$$

<sup>†</sup>*Historical Footnote:* The method of variation of parameters was invented by Joseph Lagrange in 1774.